W3 – Equations of Planes Unit 6
MCV4U
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SOLUTIONS

1) Does each point lie on the plane $3x - 5y - 6z = 12$

a) A(0, 0, –2)

LS = $3(0) - 5(0) - 6(-2)$ = 12

RS = 12

LS = RS

∴ on the plane

b) B(1, 1, –2)

LS = $3(1) - 5(1) - 6(-2)$ = 10

RS = 12

LS ≠ RS

∴ not on plane.

2) Find the *x*-, *y*-, and *z*-intercepts of each plane.

a) $2x - 3y + 6z = 12$

$2x = 12$, $x = 6$, (6, 0, 0)

$-3y = 12$, $y = -4$, (0, −4, 0)

$6z = 12$, $z = 2$, (0, 0, 2)

b) $4x - 8z = 16$

$4x = 16$, $x = 4$, (4, 0, 0)

No y-int.

$-8z = 16$, $z = -2$, (0, 0, −2)

3) Write the parametric equations of each plane given its vector equation.

a) $[x, y, z] = [3, -1, 2] + t[4, -2, 3] + s[5, 1, 0]$

$\Pi: \begin{cases} x = 3 + 4t + 5s \\ y = -1 - 2t + s \\ z = 2 + 3t \end{cases}$

b) $[x, y, z] = [0, 8, 7] + t[1, 0, -3] + s[1, -4, 7]$

$\Pi: \begin{cases} x = t + s \\ y = 8 - 4s \\ z = 7 - 3t + 7s \end{cases}$

4) Write the vector equation of a plane given its parametric equations.

a) $\pi: \begin{cases} x = 1 - 9t + 4s \\ y = 8 - 7t + s \\ z = -1 - 3t + 2s \end{cases}$

$[x, y, z] = [1, 8, -1] + t[-9, -7, -3] + s[4, 1, 2]$

b) $\pi: \begin{cases} x = 4 - t - s \\ y = 3 + 4s \\ z = 2t \end{cases}$

$[x, y, z] = [4, 3, 0] + t[-1, 0, 2] + s[-1, 4, 0]$

5) Determine the *x*-, *y*-, and *z*-intercepts of the plane $[x, y, z] = [1, -3, 2] + t[4, -3, 5] + s[-1, 7, 0]$

π: $\begin{cases} x = 1+4t-s \\ y = -3-3t+7s \\ z = 2+5t \end{cases}$

x-int:

① $0 = -3-3t+7s$
② $0 = 2+5t$

$t = -2/5$

$0 = -3-3\left(\frac{-2}{5}\right)+7s$

$3 = \frac{6}{5}+7s$

$\frac{9}{35} = s$

$x = 1+4\left(\frac{-2}{5}\right)-\frac{9}{35}$

$x = \frac{35}{35} - \frac{56}{35} - \frac{9}{35}$

$x = \frac{-30}{35} = \frac{-6}{7}$

$\left(\frac{-6}{7}, 0, 0\right)$

y-int: ① $0 = 1+4t-s$
② $0 = 2+5t$

$t = \frac{-2}{5}$

$0 = 1+4\left(\frac{-2}{5}\right)-s$

$s = \frac{5}{5} - \frac{8}{5}$

$s = \frac{-3}{5}$

$y = -3-3\left(\frac{-2}{5}\right)+7\left(\frac{-3}{5}\right)$

$y = \frac{-15}{5} + \frac{6}{5} - \frac{21}{5}$

$y = -6$

$(0, -6, 0)$

z-int: ① $0 = 1+4t-s$
② $0 = -3-3t+7s$

② × ① $0 = 7+28t-7s$
② $0 = -3-3t+7s$ +

$0 = 4+25t$

$t = \frac{-4}{25}$

$0 = 1+4\left(\frac{-4}{25}\right)-s$

$s = \frac{25}{25} - \frac{16}{25}$

$s = \frac{9}{25}$

$z = 2+5\left(\frac{-4}{25}\right)$

$z = \frac{50}{25} - \frac{20}{25}$

$z = \frac{30}{25} = \frac{6}{5}$ $\left(0, 0, \frac{6}{5}\right)$

6) Write a vector equation for each plane:

a) contains the origin; has direction vectors $\vec{a} = [2, -1, 7]$ and $\vec{b} = [3, 5, 2]$

$[x, y, z] = [0, 0, 0] + t[2, -1, 7] + s[3, 5, 2]$

b) contains the points D(1, –2, 3), E(5, –1, 8), and F(3, 9, 2)

$\vec{n}_1 = \vec{DE} = [4, 1, 5]$

$\vec{n}_2 = \vec{DF} = [2, 11, -1]$

$[x, y, z] = [1, -2, 3] + t[4, 1, 5] + s[2, 11, -1]$

c) contains the point $P_0(2, -1, 5)$; parallel to the *xy*-plane

Shortcut: $z = 5$

This is scalar form. Will learn in next section.

$\vec{m}_1 = [1,0,0]$

$\vec{m}_2 = [0,1,0]$

$[x,y,z] = [2,-1,5] + t[1,0,0] + s[0,1,0]$

d) has *y*-intercept –7; parallel to the plane defined by the parametric equations $\begin{cases} x = 7 + 3t \\ y = 6 + 2t - 5s \\ z = 1 - 8t + 3s \end{cases}$

$\vec{m}_1 = [3,2,-8]$

$\vec{m}_2 = [0,-5,3]$

$[x,y,z] = [0,-7,0] + t[3,2,-8] + s[0,-5,3]$

7) Determine the vector equation of the plane that contains the points A(2, –1, 4), B(–3, 4, 5), and C(8, –1, 6).

$\vec{m}_1 = \vec{AB} = [-5,5,1]$

$\vec{m}_2 = \vec{BC} = [11,-5,1]$

$[x,y,z] = [2,-1,4] + s[-5,5,1] + t[11,-5,1]$

8) A plane is perpendicular to $[x, y, z] = [1, -10,8] + s[1,2, -1]$ and contains the point $P(-1,4, -2)$. Determine if the point $A(7,10,16)$ is also on this plane.

$\vec{m} = \vec{PA} = [8,6,18] = 2[4,3,9]$

If A is on the plane, $\vec{PA}$ would be perpendicular to $[1,2,-1]$. Check dot product.

$\vec{PA} \cdot [1,2,-1] = [8,6,18] \cdot [1,2,-1]$

$= 2$

∴ no perpendicular; & A is NOT on the plane.

Answers:

1)a) Yes **b)** No

2)a) x-int: (6, 0, 0); y-int: (0, −4, 0); z-int: (0, 0, 2) **b)** x-int: (4, 0, 0); y-int: none; z-int: (0, 0, −2)

3)a) $\pi: \begin{cases} x = 3 + 4t + 5s \\ y = -1 - 2t + s \\ z = 2 + 3t \end{cases}$ **b)** $\pi: \begin{cases} x = t + s \\ y = 8 - 4s \\ z = 7 - 3t + 7s \end{cases}$

4)a) $[x, y, z] = [1, 8, -1] + t[-9, -7, 3] + s[4, 1, 2]$ **b)** $[x, y, z] = [4, 3, 0] + t[-1, 0, 2] + s[-1, 4, 0]$

5) x-int: $(-\frac{6}{7}, 0, 0)$; y-int: (0, −6, 0); z-int: $(0, 0, \frac{6}{5})$

6) a) $[x, y, z] = [0, 0, 0] + t[2, -1, 7] + s[3, 5, 2]$ **b)** $[x, y, z] = [1, -2, 3] + t[4, 1, 5] + s[2, 11, -1]$
c) $[x, y, z] = [2, -1, 5] + t[1, 0, 0] + s[0, 1, 0]$ **d)** $[x, y, z] = [0, -7, 0] + t[3, 2, -8] + s[0, -5, 3]$

7) $[x, y, z] = [2, -1, 4] + t[-5, 5, 1] + s[6, 0, 2]$

8) Not on the plane